

Curriculum Vitae

Yen-Ju Chen

Assistant Professor

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RESEARCH INTEREST

Perception psychology, Time perception, Vision science, Spatial-temporal information processing,
Perceptual organization, Motion perception, Psychophysics, Mathematical and computational modeling

ACADEMIC RECORD

Project Assistant	Department of Psychology	02/2026~ current
Professor	Soochow University, Taipei, Taiwan	
	Graduate School of Informatics	
PhD	Department of Intelligence Science and Technology, Kyoto University, Kyoto, Japan	10/2022~ 09/2025
	Department of Psychology	
Master of Science	National Cheng Kung University, Tainan, Taiwan	09/2020 ~ 06/2022
	Department of Psychology	
Bachelor of Science	Soochow University, Taipei, Taiwan	09/2016 ~ 06/2020

RESEARCH EXPERIENCE

Visiting Scholar	School of Psychology	08/2024 ~ 09/2024
	University of Nottingham, Nottingham, UK	
	Graduate School of Informatics	
Part-time Research	Department of Intelligence Science and	10/2022 ~ 09/2025
Assistant	Technology, Kyoto University, Kyoto, Japan	
	Department of Psychology	
Part-time Research	Soochow University, Taipei, Taiwan	09/2017 ~ 09/2018
Assistant		

TEACHING & LECTURE

Lecture	Department of Law	
	Soochow University, Taipei, Taiwan	2026/02 ~ 2026/07
	<u>Course: General Psychology</u>	
Lecture	Department of Psychology	
	Soochow University, Taipei, Taiwan	2026/02 ~ 2026/07
	<u>Course: Data Analysis with R</u>	
Teaching Assistant	Department of Psychology	
	National Cheng Kung University, Tainan, Taiwan	2021/02 ~ 2021/07
	Course: Sensation and Perception	
Teaching Assistant	Department of Psychology	
	Soochow University, Taipei, Taiwan	2019/02 ~ 2020/02
	Course: Statistics for psychology and education	

PUBLICATION

Journal Article

- Chen, Y.J.**, Sun, Z., & Nishida, S. (2025). Revealing temporal dynamics of the visuomotor system via continuous tracking of position and attribute. *Journal of Vision*, 25(8), 19. <https://doi.org/10.1167/jov.25.8.19> (SCI, IF = 2.0, 2023; Ranking: 39/95 = Top 41% in Ophthalmology)
- Chen, Y.J.**, Sun, Z., & Nishida, S. (2025). Human visual grouping based on within- and cross-area temporal correlations. *PLOS Computational Biology* 21(9): e1013001. <https://doi.org/10.1371/journal.pcbi.1013001> (SCI, IF = 3.8, 2023; Ranking: 11/66=0.1 in Mathematical & Computational biology)
- Sun, Z., **Chen, Y.J.**, & Yang, Y.H., Nishida, S. (2025). Machine Learning Modeling for Multi-order Human Visual Motion Processing. *Nat Mach Intell.* <https://doi.org/10.1038/s42256-025-01068-w>. (SCI, IF = 23.9, 2024; Ranking: 1/170 = Top 0.5% in Computer science, interdisciplinary applications)
- Chen, Y.J.**, Sun, Z., & Nishida, S. (2024). Feature-invariant processing of spatial segregation based on temporal asynchrony. *Journal of Vision*, 24(5), 15. <https://doi.org/10.1167/jov.24.5.15>. (SCI, IF = 2.0, 2023; Ranking: 39/95=0.4 in Ophthalmology)

Conference Article

- Li, Y., Sun, Z., **Chen, Y.J.**, Nishida, S. (2025). [Poster] Building Reasonable Inference for Vision-Language Models in Blind Image Quality Assessment. 32nd International Conference on Neural Information Processing (ICONIP), Okinawa, Japan.
- Sun, Z., **Chen, Y.J.**, Yang, Y.H., Nishida, S. (2023) Modeling Human Visual Motion Processing with Trainable Motion Energy Sensing and a Self-attention Network. 2023 Conference on Neural Information Processing Systems, New Orleans, United States. <https://doi.org/10.48550/arXiv.2305.09156>

- Sun, Z., **Chen, Y.J.**, Yang, Y.H., Nishida, S. (2023) Comparative Analysis of Visual Motion Perception: Computer Vision Models versus Human Vision. Conference on Cognitive Computational Neuroscience, Oxford, UK. https://2023.ccneuro.org/view_paper32e1.html?PaperNum=1080
- Wang, S. T., **Chen, Y. J.**, Xu, Y. D., Ting, T. T., & Chan, T. C. (2019). Application of Weighted Alternating Least Squares on Constructing the Disease Networks in the Heterogeneous Process of Aging. In 2019 20th IEEE International Conference on Mobile Data Management (MDM) pp. 513-520. IEEE. <http://doi.org/10.1109/MDM.2019.00112>

Conference Abstract

- He, Z., **Chen, Y.J.**, Yang, Y.H., Nishida, S. (2026). [poster] Multi-Layered Hierarchical Motion: Ancestors Matter, Descendants Do Not. *Annual meeting of vision science society*, Florida, USA.
- Chen, Y. J.**, Sun, Z., & Nishida, S. Y. (2025). [Poster] Temporal dynamics of spatial segmentation based on temporal correlation measured by continuous tracking. *Journal of Vision*, 25(9), 1996-1996. <https://doi.org/10.1167/jov.25.9.1996>
- Chen, Y.J.**, Sun, Z., Nishida, S. Y. (2024). [poster]. Square versus triangular wave modulation in temporal asynchrony segmentation: an insight into underlying computation. Asia Pacific Conference on Vision, Singapore.
- Chen, Y.J.**, Sun, Z., Nishida, S. Y. (2024). [poster]. The Role of Cross-Area and Within-Area Temporal Correlations in Visual Segmentation. European Conference on Visual Perception 2024. Aberdeen, UK.
- Chen, Y.J.**, Sun, Z., Nishida, S. Y. (2024). [poster]. Temporal dynamics gap between position tracking and attribute tracking. *Journal of Vision*, 24(10), 465-465. <https://doi.org/10.1167/jov.24.10.465>
- Sun, Z., **Chen, Y.J.**, Yang, Y.H., Nishida, S. Y. (2024). [oral]. Acquisition of second-order motion perception by learning to recognize the motion of objects made by non-diffusive materials. *Journal of Vision*, 24(10), 4894-4894. <https://doi.org/10.1167/jov.23.9.4894>
- Chen, Y. J.**, & Nishida, S. Y. (2023). [poster]. Temporal limits of visual segmentation based on temporal asynchrony in luminance, color, motion direction, and their mixtures. *Journal of Vision*, 23(9), 4864-4864. <https://doi.org/10.1167/jov.23.9.4864>
- Sun, Z., **Chen, Y. J.**, Yang, Y. H., & Nishida, S. Y. (2023). [poster]. Modeling of Human Motion Perception Mechanism: A Simulation based on Deep Neural Network and Attention Transformer. *Journal of Vision*, 23(9), 4894-4894. <https://doi.org/10.1167/jov.23.9.4894>
- Chen, Y. J.**, & Huang, P. C. (2022). [poster]. Generalized form of common fate: evidence of structure dominance effect. *Journal of Vision*, 22(14), 3830-3830. <https://doi.org/10.1167/jov.22.14.3830>
- Kao, J. F., **Chen, Y. J.**, Huang, P. C., & Hsieh, S. (2022). [poster]. How age affects the preparation effects under cross-modal switching: Evidence from behavioral and ERP measures. In *PERCEPTION* (Vol. 51, pp. 157-157). 1 OLIVERS YARD, 55 CITY ROAD, LONDON EC1Y 1SP, ENGLAND: SAGE PUBLICATIONS LTD.
- Chen, Y. J.**, & Huang, P. C. (2021). [poster]. Temporal synchrony accompany with structure cue is more effective in the segmentation task. *Journal of Vision*, 21(9), 2137-2137. <https://doi.org/10.1167/jov.21.9.2137>
- Chen, Y.J.**, Huang, P.C. (2021). [oral]. Temporal synchrony or structure? A potential computational framework, 2021 Taiwan Psychology Affair, Taipei, Taiwan.

Chen, Y.J., Wang, M.Y. (2019). [poster]. Influence of Arousal to time perception under positive IAPS picture: Controlling confound of approach motivation. 2019 Taiwan Psychology Affair, Kaohsiung, Taiwan.

Preprint

Li, Y., Sun, Z., **Chen, Y.J.**, Nishida, S. (2025). Investigate the Low-level Visual Perception in Vision-Language based Image Quality Assessment [Preprint]. *ArXiv*. <http://doi.org/10.48550/arXiv.2512.09573>

Thesis

Chen, Y.J. (2025) (Doctor Dissertation). Computing temporal correlation for perceptual organization in vision. Supervisor: Shin'ya Nishida.

Chen, Y.J. (2022) (Master Thesis) Temporal Structure does not trump synchrony in low temporal frequency: Implication and its potential interpretation. Supervisor: Pi-Chun Huang.

GRANTS

From Perceptual Structure to Semantic Space: A Human-centered Computational Framework for Object Recognition

- Period: April 2026 to March 2029
- Grant source: National Science and Technology Council, Taiwan (NSTC)
- Grant sequence: NSTC 115-2410-H-031-011-MY3
- Grant amount: TWD 4,608,000

Constructing a human-like dynamic perceptual organization model: A visual transformer basic approach

- Period: March 2023 to September 2025
- Grant source: Kyoto University DoGS SPRING program, Japan Science and Technology Agency, Japan (JST) , Kyoto University Science and Technology Innovation Creation Fellowship
- Grant sequence: JPMJSP2110
- Grant amount: JPY 4,890,000

Time flies: the influence of arousal and approach motivation on time perception

- Period: August 2019 to July 2020
- Grant source: College Student Research Scholarship, National Science and Technology Council, Taiwan (NSTC)
- Grant sequence: 108-2813-C-031-010-H
- Grant amount: NTD 48,000

AWARDS

College Student Research Creativity Award

- March 2020

- Ministry of Science and Technology, Taiwan

PROFESSIONAL AFFILIATIONS

Member, <i>Taiwan Psychology Association</i> , Taipei, Taiwan	2021-2022
Member, <i>Vision Science Society</i> , Florida, US	2021-2026